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plate has many pores, it is called **multiple perforation plate**. Example *Liriodendron*.

The secondary wall thickening of vessels are annular, spiral, scalariform, reticulate, or pitted as in tracheids, Vessels are chief water conducting elements in Angiosperms and absent in Pteridophytes and Gymnosperms. In *Gnetum* of Gymnosperm, vessels occur. The main function is conduction of water, minerals and also offers mechanical strength.

Xylem Fibre

The fibres of sclerenchyma associated with the xylem are known as xylem fibres. Xylem fibres are dead cells and have lignified walls with narrow lumen. They cannot conduct water but being stronger provide mechanical strength. They are present in both primary and secondary xylem. Xylem fibres are also called libriform fibres.

The fibres are abundantly found in many plants. They occur in patches, in continuous bands and sometimes singly among other cells. Between fibres and normal tracheids, there are many transitional forms which are neither typical fibres nor typical tracheids. The transitional types are designated as **fibre-tracheids**. The pits of fibre-tracheids are smaller than those of vessels and typical tracheids.

Vessels are found in Gymnosperms like *Ephedra, Gnetum* and *Welwitschia*

Vesselless angiospermic families *Winteraceae, Tetracentraceae* and *Trochodendracae*.

Xylem Parernchyma

The parenchyma cells associated with the xylem are known as xylem parenchyma. These are the only living cells in xylem tissue. The cell wall is thin and made up of cellulose. Parenchyma arranged longitudinally along the long axis is called **axial parenchyma**. Ray parenchyma is arranged in radial rows. Secondary xylem consists of both axial and ray parenchyma, Parenchyma stores food materials and also helps in conduction of water.

Phloem

Phloem is the food conducting complex tissues of vascular plants. The term phloem was coined by **C. Nageli** (1858) The Phloem which is derived from procambium is called primary phloem and the phloem which is derived from vascular cambium is called secondary phloem. Early formed primary phloem elements are called **protophloem** whereas the later formed primary phloem elements are called **metaphloem**. Protophloem is short lived. It gets crushed by the developing metaphloem.

Phloem Consists of Four Types of Cells

1. Sieve elements
2. Companion cells
3. Phloem parenchyma
4. Phloem fibres

Sieve Elements

Sieve elements are the conducting elements of the phloem. They are of two types, namely sieve cells and sieve tubes.

Sieve Cells

These are primitive type of conducting